

## Week 5 - Class 10: Loops: while, do-while, for

### Why This Matters

Without loops, you'd repeat code manually. Loops let you:

- Repeat actions efficiently.
- Process unknown amounts of data.
- Traverse sequences.

C++ provides while, do-while, and for loops.

### while Loop

Repeats while condition is true.

```
int i = 0;
while (i < 5) {
    cout << i << "\n";
    i++;
}
```

### do-while Loop

Executes body once, then repeats while condition is true.

```
int x;
do {
    cout << "Enter positive number: ";
    cin >> x;
} while (x <= 0);
```

### for Loop

Provides init, condition, increment in one line.

```
for (int i = 1; i <= 5; i++) {
    cout << "5 * " << i << " = " << 5*i << "\n";
}
```

### Infinite Loops

Occur when condition never becomes false. Often used with break to exit manually.

```
while (true) {
    cout << "Forever!\n";
    break;
}
```

### break and continue

break exits loop immediately. continue skips rest of current iteration.

```
for (int i=0;i<10;i++){
    if (i==5) continue;
    if (i==8) break;
    cout<<i<<"\n";
}
```

### Loop Flow Diagrams

While: check condition -> body -> repeat

Do-while: body -> check condition -> repeat

For: init -> condition -> body -> increment -> condition...

### Common Mistakes

1. Forgetting to update loop variable -> infinite loop.
2. Using = instead of == in conditions.
3. Off-by-one errors (<= vs <).

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4. Thinking do-while is identical to while (it runs at least once).

### Exercises

1. Countdown 10 to 1 with while, then print Liftoff!
2. Password loop with do-while until correct.
3. For loop printing squares 1..10.
4. Skip even numbers with continue.
5. Stop on input 0 with break.

### Project Tie-In

Expand Control Flow Demo into Menu Loop:

- while(true) show menu.
- switch(choice) handle options.
- break to quit.

This creates an interactive menu system.